Technical Report: Simulation-Based Handover Decision Evaluation for High-Mobility LTE Users

Project Title: LostPacket

Team ID: A3-PS016-TC243

Problem Statement: PS-016 — Simulation-Based Handover Decision Evaluation for High-Mobility Users

Category: 6G & Future Networks

Simulator: ns-3.48 (Discrete-Event Network Simulator, LTE/EPC Module Only)

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Abstract

High-mobility vehicular communication corridors present severe operational challenges for cellular handovers, including late triggering, radio link failure, packet loss, and ping-pong oscillations. This study presents LostPacket, a fully reproducible, standards-compliant simulation and machine learning evaluation framework developed for ns-3.48 LTE/EPC. We deploy a three-cell linear LTE network corridor traversed by a high-velocity User Equipment (UE) operating at 30 km/h (low mobility) and 120 km/h (high mobility). We benchmark the classical 3GPP Event A3 Reference Signal Received Power (RSRP) heuristic against an autonomous supervised Random Forest controller implemented in native C++ and integrated directly into the ns-3 LTE Handover Management Service Access Point (SAP). To ensure scientific integrity, strict seed isolation was enforced: 1,685 genuine measurement samples were collected across seeds 1–5 for imitation learning, while a 20-run evaluation matrix was conducted on held-out seeds 6–10. Bit-level inference parity between Python scikit-learn and the native C++ engine was verified (0.0000000000×10^0 error). Across all 20 held-out evaluation runs, both policies achieved 100% handover success (zero failures), zero ping-pong events, sustained downlink application throughput above 0.953 Mbps (99.46%–99.95% packet delivery ratio), and limited the handover-associated downlink packet reception gap to 31–60 ms. The Random Forest controller successfully imitated the Event A3 policy with 100% precision and recall on unseen seeds, proving that learned decision policies can be deterministically embedded within discrete-event network simulation kernels.

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1. Problem Statement

In cellular radio access networks, handovers transfer an ongoing data session between base stations (eNodeBs) to maintain connectivity as users move through space. High-velocity transit (such as highway vehicular mobility at 120 km/h) creates acute challenges:

- **Steep Path Loss Gradients**: The User Equipment (UE) traverses cell boundaries rapidly, experiencing sudden signal decay.

- **Doppler Shift and Multipath Distortion**: Channel conditions fluctuate rapidly, compressing the available decision window.

- **Too-Late Handover Dilemma**: Conservative triggering thresholds or prolonged Time-to-Trigger (TTT) values delay execution until signal quality drops below the receiver sensitivity threshold, causing radio link failure (RLF), call drop, and severe user-plane blackout.

- **Too-Early / Ping-Pong Oscillation**: Overly aggressive triggering thresholds cause premature handovers and rapid bouncing ($A \to B \to A$) between adjacent base stations, flooding the core network with control-plane signaling and inducing repetitive user-plane interruptions.

Addressing this trade-off requires evaluating handover decision algorithms under controlled, reproducible conditions using discrete-event network simulation.

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2. Objectives

The primary engineering and scientific objectives of this investigation are:

1. Construct an explicit, standards-compliant three-cell LTE simulation environment using ns-3.48 without 5G abstractions.

2. Model vehicular mobility across low-mobility (30 km/h) and high-mobility (120 km/h) conditions.

3. Establish a 3GPP Event A3 baseline configured with standard hysteresis and Time-to-Trigger parameters.

4. Instrument comprehensive packet-level and control-plane metrics: handover attempts, successes, failures across all four eNodeB failure causes, ping-pong events, throughput, packet delivery ratio, and packet reception gaps.

5. Collect genuine simulation data without synthetic labels to train a supervised Random Forest classifier via imitation learning.

6. Integrate the trained Random Forest directly into the ns-3 C++ Handover Management SAP so that

machine learning inferences actively govern simulation execution.

7. Conduct a statistically sound, held-out evaluation across multiple random seeds (seeds 6–10) to quantify policy trade-offs and verify operational stability.

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3. System Architecture

The LostPacket system architecture couples the ns-3.48 discrete-event network simulator with a native C++ machine learning decision engine. The network consists of:

- **Evolved Packet Core (EPC)**: Instantiated via `ns3::PointToPointEpcHelper`, modeling the Packet Data Network Gateway (PGW), Serving Gateway (SGW), and Mobility Management Entity (MME).

- **Remote Internet Host**: Connected to the PGW via a Point-to-Point link (100 Gbps bandwidth, 10 ms propagation delay).

- **Three LTE Macro eNodeBs**: Deployed in a linear corridor with inter-eNodeB X2 communication interfaces.

- **Vehicular User Equipment (UE)**: Mobile node attached to the EPC and running a downlink UDP sink application.

- **Handover Management SAP**: Interface connecting the eNodeB Radio Resource Control (RRC) entity to the handover algorithm.

Figure 1: LostPacket LTE Handover Simulation Architecture

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4. LTE Simulation Configuration

4.1 Physical Radio and Spectrum Configuration

All physical layer and propagation parameters are explicitly configured to ensure reproducibility:

- **Spectrum Channel**: `MultiModelSpectrumChannel` providing independent channel realizations.

- **Propagation Model**: Explicit `ns3::FriisPropagationLossModel` with free-space path loss attenuation.

- **eNodeB Transmission Power**: 46.0 dBm (40 W per carrier).

- **Antenna Model**: `IsotropicAntennaModel` installed on both eNodeB and UE nodes.

- **Carrier Frequency**: Downlink EARFCN 100 (2120 MHz); Uplink EARFCN 19100 (1930 MHz).

- **Bandwidth**: 25 Resource Blocks (5 MHz bandwidth).

- **MAC Scheduler**: Round-Robin scheduling (`ns3::RrFfMacScheduler`).

4.2 Network Topology, Trajectory, and Observation Windows

The network topology comprises three linear eNodeBs:

- eNodeB 1: $x = 0\text{ m}, y = 0\text{ m}, z = 0\text{ m}$

- eNodeB 2: $x = 500\text{ m}, y = 0\text{ m}, z = 0\text{ m}$

- eNodeB 3: $x = 1000\text{ m}, y = 0\text{ m}, z = 0\text{ m}$

The UE starts at coordinate $x = -100\text{ m}, y = 0\text{ m}, z = 0\text{ m}$ and travels in the positive x-direction at constant velocity:

- **Speed Class 1 (Low Mobility)**: $30\text{ km/h} \approx 8.333\text{ m/s}$.

- **Speed Class 2 (High Mobility)**: $120\text{ km/h} \approx 33.333\text{ m/s}$.

Trajectory and Observation Window Clarification

To ensure scientifically defensible observation windows:

- The active multi-cell service corridor extends from $x = -100\text{ m}$ to $x = 1200\text{ m}$ (1300 m distance; 200 m past eNodeB 3). The transit time across this corridor is 156.0 s at 30 km/h and 39.0 s at 120 km/h.

- To prevent premature termination immediately following the second handover, an extended observation margin of $1.5\times$ was applied, yielding total simulation durations of 240.0 s (30 km/h) and 60.0 s (120 km/h).

- Under constant velocity, the UE covers an identical total distance of 2000.0 m (from $x = -100\text{ m}$ to $x = +1900\text{ m}$) in both scenarios.

- The terminal 700 m segment ($x = 1200\text{ m}$ to $x = 1900\text{ m}$) provides an extended post-handover observation window of 84.0 s at 30 km/h and 21.0 s at 120 km/h (well beyond the 5.0 s ping-pong detection window). This confirms that no delayed ping-pong triggers or unhandled link failures occur as the UE enters the fringe of Cell 3.

4.3 Traffic Model

Continuous downlink UDP traffic is generated by the remote host directed to port 9000 on the UE:

- Packet size: 1200 bytes.

- Packet interval: 10 ms ($100\text{ packets/second}$).

- Offered load: 0.96 Mbps continuous downlink.

- Application start time: $t = 0.1\text{ s}$; Stop time: $t = \text{simTime}$.

- Total transmitted packets: $23,990\text{ packets}$ for 240 s; $5,990\text{ packets}$ for 60 s.

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5. Baseline Handover Algorithm (3GPP Event A3)

The baseline policy implements the 3GPP Event A3 heuristic via `ns3::A3RsrpHandoverAlgorithm`. Event A3 triggers when a neighboring cell's Reference Signal Received Power ($RSRP_{target}$) exceeds the serving cell's power ($RSRP_{serving}$) by an offset margin:

$$RSRP_{target} > RSRP_{serving} + Hysteresis \quad \text{for duration } TimeToTrigger$$

Parameters configured according to 3GPP specifications:

- **Hysteresis**: 3.0 dB (prevents spurious triggering due to minor channel variations).

- **TimeToTrigger (TTT)**: 256 ms (requires sustained signal superiority before execution).

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6. Supervised Machine Learning Methodology

6.1 Imitation Learning Framework

Rather than introducing unverified reinforcement learning or synthetic rule-sets, the Random Forest model is formulated as an **imitation learning agent**. It observes real-time UE measurement reports and learns to replicate the operational boundary of the 3GPP Event A3 algorithm. High classification accuracy confirms behavioral imitation fidelity, establishing that a machine learning policy can accurately capture and execute standard cellular handovers.

6.2 Feature Space Definition

For each incoming UE measurement report, an 8-dimensional feature vector is extracted:

1. `serving_rsrp_dbm`: RSRP of the current serving cell (dBm).

2. `target_rsrp_dbm`: RSRP of the candidate neighbor cell (dBm).

3. `delta_rsrp_db`: Relative signal strength difference ($RSRP_{target} - RSRP_{serving}$).

4. `serving_rsrq_db`: Reference Signal Received Quality of serving cell (dB).

5. `target_rsrq_db`: RSRQ of candidate neighbor cell (dB).

6. `delta_rsrq_db`: Relative signal quality difference ($RSRQ_{target} - RSRQ_{serving}$).

7. `num_neighbours`: Number of visible neighbor cells reported.

8. `speed_kmh`: Velocity of the UE (30 or 120 km/h).

6.3 Dataset Provenance and Class Balance

Training data was collected across 10 simulation runs using independent random seeds 1, 2, 3, 4, and 5 at both 30 km/h and 120 km/h. A total of 1,685 measurement samples were recorded:

- **Class 0 (Hold / No Handover)**: 1,665 samples (98.81%).

- **Class 1 (Trigger Handover)**: 20 samples (1.19%).

This acute class imbalance reflects real-world cellular environments, where handovers are brief transition events separated by long periods of steady tracking. To prevent majority-class bias, balanced class weighting was applied during training ($w_c = N / (2 \cdot N_c)$).

6.4 Model Architecture and Feature Importance

The Random Forest ensemble was trained with the following hyperparameter specification:

- Number of estimators (M): 30 decision trees.

- Maximum tree depth: 5.

- Minimum samples per split: 4.

- Splitting criterion: Gini impurity.

- Class weight: `balanced`.

- Random state seed: 42.

Feature importance analysis confirms that differential metrics govern the decision:

1. `delta_rsrq_db`: 0.2975

2. `serving_rsrq_db`: 0.2336

3. `delta_rsrp_db`: 0.2224

4. `target_rsrq_db`: 0.1382

5. `target_rsrp_dbm`: 0.0628

6. `serving_rsrp_dbm`: 0.0362

7. `speed_kmh`: 0.0093

8. `num_neighbours`: 0.0000

Figure 2: Random Forest Feature Importance Distribution

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7. Training and Evaluation Methodology

7.1 Autocorrelation and Seed-Level Separation

In cellular simulation, successive measurement reports received every 240 ms from a moving vehicle exhibit high autocorrelation. Shuffling samples across time within the same simulation run induces temporal data leakage.

To ensure complete statistical independence:

- **Training Set**: Collected strictly from Seeds 1, 2, 3, 4, 5 (1,685 samples).

- **Held-Out Evaluation Set**: Collected independently from Seeds 6, 7, 8, 9, 10 (1,685 samples). No samples from seeds 6–10 were accessible during model fitting.

7.2 Training Cross-Validation

Across 5-fold stratified cross-validation on seeds 1–5, the model achieved:

- Precision: 1.0000 ± 0.0000

- Recall: 1.0000 ± 0.0000

- F1-Score: 1.0000 ± 0.0000

- ROC-AUC: 1.0000 ± 0.0000

Figure 3: Training Confusion Matrix (Seeds 1 to 5 Cross-Validation)

7.3 Held-Out Test Dataset Evaluation (Seeds 6 to 10)

When evaluated offline on the 1,685 samples collected from held-out seeds 6–10 (containing 1,665 Hold and 20 Handover ground-truth events):

- Test Accuracy: 1.0000

- Test Precision: 1.0000

- Test Recall: 1.0000

- Test F1-Score: 1.0000

- Confusion Matrix: `[[1665, 0], [0, 20]]`

Figure 4: Held-Out Test Confusion Matrix (Seeds 6 to 10)

This demonstrates complete behavioral fidelity in learning the A3 triggering boundary across unseen simulation seeds.

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8. Model Integration and Controller Architecture

8.1 Native C++ Tree Traversal Engine

To avoid external inter-process communication (IPC) overhead, socket latency, or Python GIL contention during discrete-event execution, the trained forest was serialized into portable JSON (`model/rf_model.json`) and compiled directly into a C++ header (`scratch/rf-model-generated.h`).

The native C++ inference engine traverses 30 trees in memory. Exact probability parity between Python scikit-learn and the C++ engine was verified across all samples:

$$\max |P_{\text{sklearn}}(y=1|X) - P_{\text{C++}}(y=1|X)| = 0.0000000000 \times 10^0$$

8.2 Data Flow and Pending Handover Guard

The `RfHandoverAlgorithm` class subclasses `ns3::LteHandoverAlgorithm`. When a measurement report arrives via `DoReportUeMeas`, the controller:

1. Filters candidate neighbor cells and extracts the 8-dimensional feature vector.

2. Checks whether a handover was initiated for the same UE within a 2.0 s guard interval. If active, the report is suppressed to eliminate command duplication.

3. Computes the ensemble probability $P(y=1|X)$. If $P \ge 0.50$, it dispatches `m_handoverManagementSapUser->TriggerHandover(rnti, targetCellId)`.

Figure 5: Data Flow Diagram of Random Forest Decision Pipeline

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9. Experimental Setup and Multi-Seed Matrix

The experimental matrix comprises 20 independent simulation runs conducted on held-out seeds 6 through 10:

- **Policies**: 3GPP Event A3 Baseline vs. Native Random Forest Controller.

- **Mobility Speeds**: 30 km/h (240 s duration) and 120 km/h (60 s duration).

- **Seeds**: 6, 7, 8, 9, 10.

- **Traffic**: Uniform downlink UDP stream (1200 B / 10 ms = 0.96 Mbps).

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10. Metrics and Mathematical Definitions

1. **Handover Attempts (N_{att})**: Number of handover executions initiated by the controller via the Handover Management SAP.

2. **Handover Successes (N_{succ})**: Successfully completed handovers verified by the target eNodeB RRC callback `HandoverEndOk`.

3. **Handover Failures (N_{fail})**: Deduplicated control-plane failures tracked via eNodeB callbacks (`HandoverFailureNoPreamble`, `HandoverFailureMaxRach`, `HandoverFailureLeaving`,

`HandoverFailureJoining`) and UE callback `HandoverEndError`.

4. **Ping-Pong Events (N_{pp})**: Defined as a handover sequence from Cell A to Cell B followed by a return handover to Cell A ($A \to B \to A$) within a observation window of $\Delta t \le 5.0\text{ s}$.

5. **Throughput (Th)**: Average downlink application throughput calculated from received packets:

$$Th = \frac{N_{rx_bytes} \cdot 8}{t_{sim} \cdot 10^6} \quad [\text{Mbps}]$$

6. **Packet Delivery Ratio (PDR)**: Ratio of received packets to transmitted packets traced directly at the application boundary:

$$\text{PDR} = \frac{N_{rx_packets}}{N_{tx_packets}} \times 100\%$$

7. **Handover-Associated DL Packet Reception Gap**: Time interval elapsed between the last downlink packet received before handover initiation and the first packet received after target cell attachment:

$$\Delta t_{gap} = t_{first_post_rx} - t_{last_pre_rx} \quad [\text{seconds}]$$

Note: This metric is explicitly defined as an application-layer operational proxy rather than a physical-layer hardware interruption measurement.

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11. Results and Visualizations

Across all 20 held-out evaluation runs, every experiment completed successfully with valid per-run CSV traces.

11.1 Handover Control-Plane Performance Summary

Policy	Speed (km/h)	HO Attempts	HO Successes	HO Failures	Ping-Pong Events	Mean DL Reception Gap (s)
A3	30	2.00 +/- 0.00	2.00 +/- 0.00	0.00 +/- 0.00	0.00 +/- 0.00	0.0340 +/- 0.0338
A3	120	2.00 +/- 0.00	2.00 +/- 0.00	0.00 +/- 0.00	0.00 +/- 0.00	0.0600 +/- 0.0156
RF	30	2.00 +/- 0.00	2.00 +/- 0.00	0.00 +/- 0.00	0.00 +/- 0.00	0.0310 +/- 0.0305
RF	120	2.00 +/-	2.00 +/- 0.00	0.00 +/-	0.00 +/- 0.00	0.0580 +/- 0.0175

*	0.00	0.00

11.2 Downlink User-Plane Data Delivery Summary

Policy	Speed (km/h)	TX Packets	RX Packets	Packet Loss	PDR (%)	Throughput (Mbps)
A3	30	23990 +/- 0	23975.2 +/- 5.8	14.8 +/- 5.8	99.938 +/- 0.024	0.9590 +/- 0.0002
A3	120	5990 +/- 0	5966.8 +/- 7.3	23.2 +/- 7.3	99.613 +/- 0.122	0.9547 +/- 0.0012
RF	30	23990 +/- 0	23976.8 +/- 5.5	13.2 +/- 5.5	99.945 +/- 0.023	0.9591 +/- 0.0002
RF	120	5990 +/- 0	5957.4 +/- 4.9	32.6 +/- 4.9	99.456 +/- 0.082	0.9532 +/- 0.0008

Figure 6: Downlink Application Throughput Comparison

Figure 7: Packet Delivery Ratio Comparison

Figure 8: Handover-Associated DL Packet Reception Gap Comparison

Figure 9: Time-Series Throughput Dynamics at 120 km/h

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12. Discussion

1. **Handover Reliability**:

Both policies achieved 100% handover success across all seeds and speeds. In every run, exactly two handovers occurred ($1 \to 2$ and $2 \to 3$), matching the corridor geometry. Zero handover failures were recorded.

2. **Ping-Pong Suppression**:

Zero ping-pong events occurred under the 5.0-second window. The combination of directional velocity, Friis path loss progression, and the 2.0-second pending handover guard interval prevented oscillations.

3. **Throughput and Packet Delivery**:

Downlink application throughput was sustained near the theoretical 0.96 Mbps maximum (0.959 Mbps at 30 km/h; 0.953–0.954 Mbps at 120 km/h). Packet delivery ratios exceeded 99.45% under high-speed conditions.

4. **Packet Reception Gap Dynamics**:

The Random Forest controller exhibited slightly shorter mean packet reception gaps (31.0 ms vs 34.0 ms at 30 km/h; 58.0 ms vs 60.0 ms at 120 km/h). The Random Forest evaluates incoming measurement reports every 240 ms and triggers execution immediately once posterior probability exceeds threshold, whereas Event A3 waits for its continuous TimeToTrigger timer.

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13. Technical Limitations

1. **Application-Layer Reception Gap Proxy**:

The measured packet reception gap represents the inter-arrival latency at the UDP application layer. It serves as an operational proxy rather than a physical-layer user-plane interruption duration, as it incorporates scheduling latency and buffer draining delays.

2. **Imitation Learning Scope**:

The Random Forest model was trained on ground-truth decisions generated by the 3GPP Event A3 baseline. Consequently, high accuracy reflects behavioral imitation rather than unconstrained policy superiority.

3. **Propagation Environment**:

The simulation uses the standard free-space Friis propagation loss model. Urban canyon multipath fading, shadow fading, and building blockages were not simulated.

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14. Conclusion

This investigation delivered an end-to-end, reproducible cellular handover evaluation framework in ns-3.48 for Problem Statement PS-016. By implementing a native C++ Random Forest controller subclassing `ns3::LteHandoverAlgorithm` with bit-level parity to scikit-learn, the framework demonstrated that machine learning policies can reliably govern discrete-event handover decisions. Benchmarking across 20 independent simulation runs on strictly held-out seeds confirmed that the learned policy faithfully captures the stability and reliability of the 3GPP Event A3 standard, sustaining 99.46%–99.95% packet delivery ratios with zero failures and zero ping-pongs up to $120\ text{ km/h}$.

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15. Reproducibility Instructions

All experiments and deliverables can be reproduced from the project directory:

```
# 1. Build ns-3.48 simulator

cd /home/arjya/projects/lostpacket/ns-3.48

./ns3 build



# 2. Collect genuine training data across seeds 1-5

./scripts/collect_training_data.sh



# 3. Train Random Forest model, verify inference parity, and export headers

python3 scripts/train_rf.py
```

```
# 4. Generate architectural and data flow diagrams

python3 scripts/generate_diagrams.py




# 5. Collect held-out evaluation dataset across seeds 6-10

./scripts/collect_heldout_data.sh




# 6. Evaluate Random Forest model offline on held-out seeds

python3 scripts/eval_rf_heldout.py




# 7. Execute 20-run held-out evaluation matrix (seeds 6-10)

./scripts/run_evaluation.sh




# 8. Compute aggregate statistics and generate KPI figures

python3 scripts/analyze_results.py




# 9. Recompile PDF Technical Report

python3 scripts/generate_pdf_report.py
```

```
# 10. Run live demonstration from root

cd /home/arjya/projects/lostpacket

./demo.sh
```

References

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